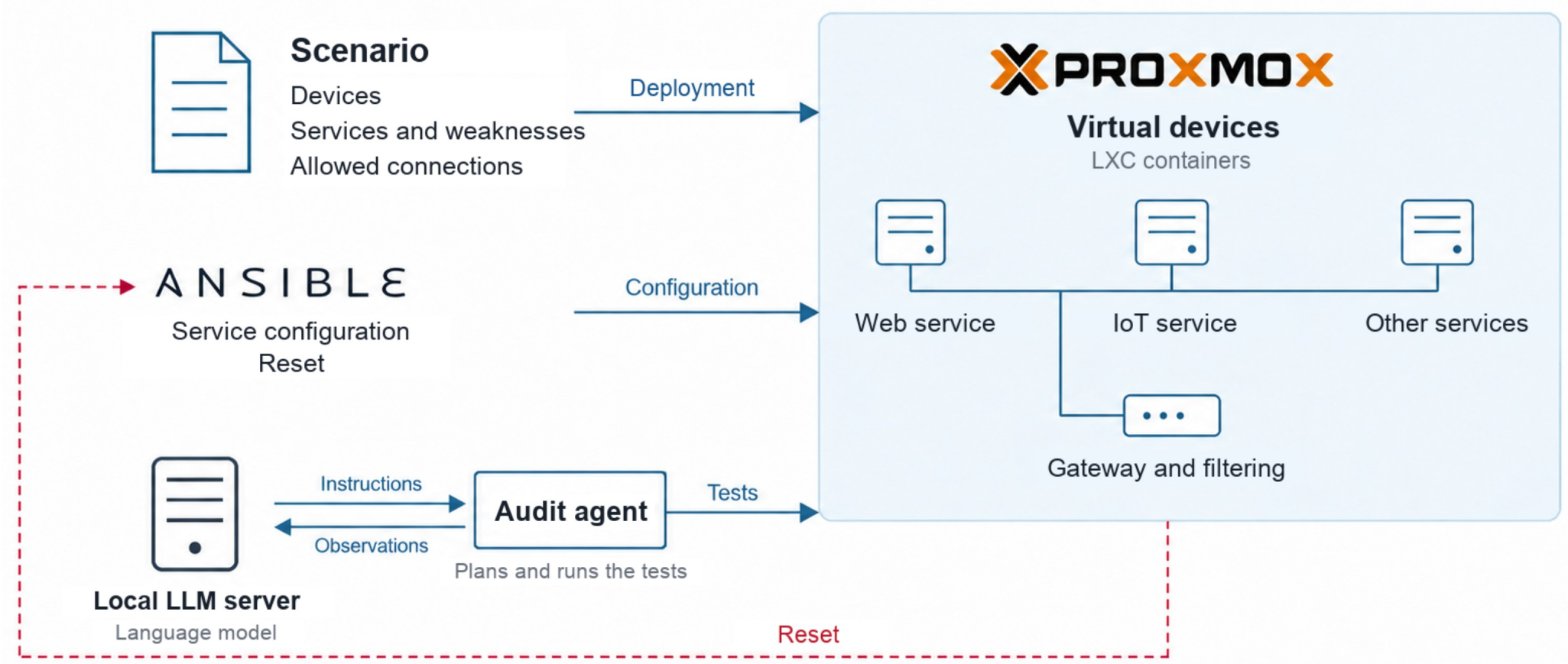
SMART'ILIA
UMONS

LANCE

An autonomous and verifiable auditor for IoT/IIoT networks

23 October 2026 · Université de Mons, ILIA

Architecture: deploy, test, reset



Two parts: benchmark and sandbox

Benchmark

and evaluation

Scenarios with known weaknesses; every finding is scored against them

Sandbox

for models and scripts

Launch a local model or a script on the same scenarios and keep the evidence

V1: 12 scenarios in six families

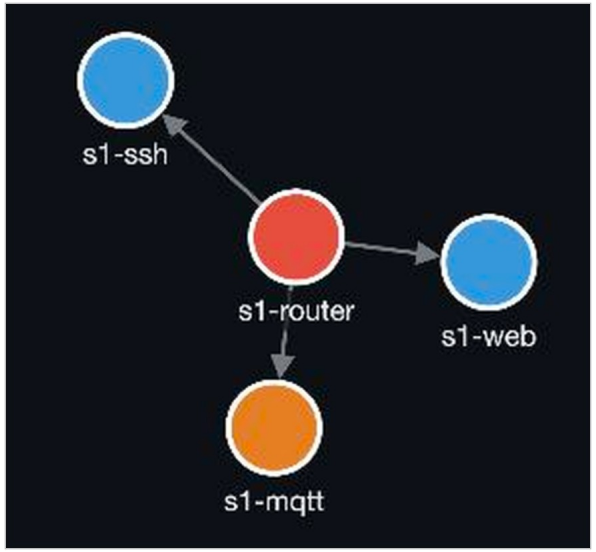
Organisation	Scenarios	Devices	Weaknesses	Access to examine
Flat network	S1, S2, S9, S10	22	49	no imposed relay
Gateway	S3	8	18	internal services after a pivot
Segmented zones	S4, S8	16	32	filtered communication between zones
Star network	S5, S6	14	31	exchanges through a central device
Local–cloud	S7	6	14	from a local service to a remote one
Scale	S11, S12	50	65	networks of 15 and 35 devices
Total	12	116	209	

V2: families vary by device, architecture and weakness

Category	What varies	Example variants
Devices	the device under test	MQTT formats, gateway, camera, sensor
Architecture	how the devices are connected	flat, segmented zones, gateway, star
Weaknesses	the type of flaw to find	credentials, data exposure, injection, crypto

Examples: S1 and S15

S1: flat network



router and 3 services, one subnet; 12 expected findings

S15: authenticated multi-tenant API

a low-privilege tenant token leads to a cross-tenant administrative export, through three flaws in the API; 3 expected findings

Scenario	TP	FP	F1	Precision	Recall
S1	12	6	80 %	66.7 %	100 %
S15	3	5	54.5 %	37.5 %	100 %

Example evidence, S1

MQTT broker on port 1883 accepts anonymous connections: a subscription to # without credentials returns retained admin topics. Status: confirmed.

Sandbox: launch a model on a scenario

Scenario
same network for every run

→

Local model
Qwen3.8 27B on our servers
context about 50k tokens

→

Findings
scored against the answer key

Four launch modes on the same scenario

Mode	What runs
LANCE pipeline	phased agent: analysis per device, one test per weakness
Rule-based audit	fixed rules, scored
Simple agent	a model alone, in a bounded loop
Script	a deterministic procedure, unscored baseline

Next steps

1 · Structure the benchmark better

Organise the scenario families by devices, architecture and weaknesses, within each V2 category

2 · Improve the sandbox

Make the launch modes easier to run and compare

Simple agent	**Rule-based audit**	**Script**
a model alone	fixed rules	a fixed procedure

3 · Submit in 2026